In [1]:
```
#Let consider   3x+2y=12
#               2x-y=1    find x and y
#Method-1
```

In [3]:
```
import numpy as np
a=np.array([[3,2],[2,-1]])
b=np.array([12,1])
print(a)
print(b)
```

```
[[ 3  2]
 [ 2 -1]]
[12  1]
```

In [4]:
```
x=np.linalg.solve(a,b)
```

In [5]:
```
print("Solution=",x)
```

```
Solution= [2. 3.]
```

In [6]:
```
#Let consider   3x+2y=12
#               2x-y=1    find x and y
#Method-2
```

In [7]:
```
a=np.array([[3,2],[2,-1]])
b=np.array([12,1])
print(a)
print(b)
```

```
[[ 3  2]
 [ 2 -1]]
[12  1]
```

In [8]:
```
ai=np.linalg.inv(a)
```

In [9]:
```
print(ai,type(ai))
```

```
[[ 0.14285714  0.28571429]
 [ 0.28571429 -0.42857143]] <class 'numpy.ndarray'>
```

In [10]:
```
x=np.matmul(ai,b)
print("Solution=",x)
```

```
Solution= [2. 3.]
```

In [11]:
```
#OR
x=np.dot(ai,b)
print("Solution=",x)
```

```
Solution= [2. 3.]
```

In [ ]: